Midterm 2 Review Session

Contents

Configuring R 1

1 Introduction 2

2 Part 1: Kaplan-Meier and Nelson-Aalen 3
- 2.1 The risk set shrinks by the number of subjects who *leave* . . . 3
- 2.2 Nelson-Aalen estimates the *cumulative hazard* first, then exponentiates . . . 4
- 2.3 Survival estimates are *step functions*: use the most recent event time . . . 5
- 2.4 Median survival time is read off the curve, not modeled . . . 5
- 2.5 How censored times are used: denominator yes, numerator no . . . 6

3 Part 2: Cox proportional hazards models 6
- 3.1 Writing the model: name every assumption and show where it is used . . . 7
- 3.2 Interpreting a hazard ratio: magnitude, reference, adjustment, *and* significance . . . 8
- 3.3 A hazard ratio for a non-unit change: exponentiate . . . 9
- 3.4 Comparing two coefficients: variance of a *difference* . . . 9
- 3.5 A confidence interval for a hazard ratio: build it on the log scale, then exponentiate 11

4 Summary of the most common errors 11

Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
```

```r
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```r
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

1 Introduction

This chapter walks through the **most common mistakes** that students made on Midterm 2, which covered survival analysis and Cox proportional hazards models[1]. Each section states a mistake, explains *why* it is wrong, and works through the correct approach.

The exam had two parts:

- **Part 1** (Kaplan-Meier[2] and Nelson-Aalen estimators[3]) used a small data set of survival times: `10, 14, 14+, 15+, 18, 21, 25` months (the `+` marks censored[4] observations).
- **Part 2** (Cox proportional hazards regression) used a model fit to $n = 3{,}142$ men from the Western Collaborative Group Study (WCGS), relating time to incident coronary heart disease (CHD) to baseline cigarette smoking and other covariates.

[1] proportional-hazards-models.qmd#def-ph-model
[2] intro-to-survival-analysis.qmd#def-KM-estimator
[3] intro-to-survival-analysis.qmd#def-na-cuhaz-est
[4] intro-to-survival-analysis.qmd#def-right-censor

Tip

The mistakes in this chapter are organized so that the most consequential and most frequent errors come first within each part. If you only have time to review a few things, start at the top of each part.

2 Part 1: Kaplan-Meier and Nelson-Aalen

```
library(survival)
library(dplyr)

surv_data <- tibble(
  time  = c(10, 14, 14, 15, 18, 21, 25),
  death = c(1,  1,  0,  0,  1,  1,  1),
  surv  = Surv(time, death)
)
```

Here is the full table that everyone was trying to reproduce; we will refer back to it throughout this part.

```
KM_est <- survfit(surv ~ 1, type = "kaplan-meier", data = surv_data)

surv_table <-
  KM_est |>
  summary(censored = TRUE, data.frame = TRUE) |>
  as_tibble() |>
  mutate(
    hazard       = n.event / n.risk,
    nonhazard    = 1 - hazard,
    cusum_hazard = cumsum(hazard),
    surv_KM      = cumprod(nonhazard),
    surv_NA      = exp(-cusum_hazard)
  ) |>
  select(time, n.risk, n.event, hazard, surv_KM, cusum_hazard, surv_NA)

surv_table |> pander::pander()
```

Table 1: Kaplan-Meier and Nelson-Aalen calculations for the Part 1 data

time	n.risk	n.event	hazard	surv_KM	cusum_hazard	surv_NA
10	7	1	0.1429	0.8571	0.1429	0.8669
14	6	1	0.1667	0.7143	0.3095	0.7338
15	4	0	0	0.7143	0.3095	0.7338
18	3	1	0.3333	0.4762	0.6429	0.5258
21	2	1	0.5	0.2381	1.143	0.3189
25	1	1	1	0	2.143	0.1173

2.1 The risk set shrinks by the number of subjects who *leave*

Exam reference: ***Exercise 1.1*** *(6 points) — compute the KM and Nelson-Aalen estimates of* $S(t)$.

Common mistake

Reducing the number at risk[a] by **one** at every row, regardless of how many subjects actually left the study. With this mistake, the number at risk after the two censored[b] subjects (at

$t = 14^+$ and $t = 15^+$) was computed as 5 instead of 4.

[a] intro-to-survival-analysis.qmd#def-number-at-risk
[b] intro-to-survival-analysis.qmd#def-right-censor

Solution

Solution. The number at risk n_j is the number of subjects still under observation *just before* time t_j — that is, everyone who has neither had the event nor been censored yet.
Track the cohort of 7:

time	what happens	number at risk *before* this time
10	1 death	7
14	1 death **and** 1 censored	6
15	1 censored	4
18	1 death	3
21	1 death	2
25	1 death	1

At $t = 14$, one subject dies *and* a different subject is censored (the `14+`), so **two** subjects leave the risk set[a]; the count drops from 6 to 4, not from 6 to 5. Every subject who has an event **or** is censored is removed from all later risk sets.

[a] intro-to-survival-analysis.qmd#def-risk-set

2.2 Nelson-Aalen estimates the *cumulative hazard* first, then exponentiates

Exam reference: ***Exercise 1.1*** *(6 points) — compute the KM and Nelson-Aalen estimates of* $S(t)$.

⚠ Common mistake

Two versions of this error were common:

1. Computing a running sum of arbitrary quantities (e.g. summing survival probabilities, or summing $1 - \hat{\lambda}_j$) instead of summing the hazards[a] $\hat{\lambda}_j = d_j/n_j$.
2. Computing the cumulative hazard[b] correctly but then forgetting the final step, $\hat{\mathrm{S}}_{\mathrm{NA}}(t) = \exp\{-\hat{\Lambda}(t)\}$, and reporting the cumulative hazard itself (or the KM product) as the Nelson-Aalen survival estimate.

[a] intro-to-survival-analysis.qmd#def-hazard
[b] intro-to-survival-analysis.qmd#def-cuhaz

Solution

Solution. The Nelson-Aalen estimator[a] builds the **cumulative hazard** by adding up the instantaneous hazard contributions at each event time:

$$\hat{\Lambda}(t) = \sum_{j:t_j \le t} \frac{d_j}{n_j}$$

and then converts it to a survival estimate using $\mathrm{S}(t) = \exp\{-\Lambda(t)\}$:

$$\hat{\mathrm{S}}_{\mathrm{NA}}(t) = \exp\{-\hat{\Lambda}(t)\}$$

For example, at $t = 10$:

$$\hat{\Lambda}(10) = \frac{1}{7} = 0.143, \qquad \hat{\mathrm{S}}_{\mathrm{NA}}(10) = \exp\{-0.143\} = 0.867$$

Contrast this with Kaplan-Meier[b], which multiplies conditional survival probabilities[c]:

$$\hat{\mathrm{S}}_{\mathrm{KM}}(t) = \prod_{j:t_j \le t} \left(1 - \frac{d_j}{n_j}\right), \qquad \hat{\mathrm{S}}_{\mathrm{KM}}(10) = 1 - \tfrac{1}{7} = 0.857$$

The two estimators are close but **not** identical, and $\hat{\mathrm{S}}_{\mathrm{NA}}(t) \ge \hat{\mathrm{S}}_{\mathrm{KM}}(t)$ always. Reporting the same numbers for both, or reporting $\hat{\Lambda}$ where $\hat{\mathrm{S}}_{\mathrm{NA}}$ was asked for, loses credit.

[a] intro-to-survival-analysis.qmd#def-na-cuhaz-est
[b] intro-to-survival-analysis.qmd#def-KM-estimator
[c] intro-to-survival-analysis.qmd#def-kappa

2.3 Survival estimates are *step functions*: use the most recent event time

Exam reference: ***Exercise 1.3*** *(1 point) — the KM and NA survival estimates for $t = 17$ months.*

⚠ Common mistake

For $\hat{S}(17)$, looking at the row for $t = 18$ (the next event *after* 17), or interpolating between rows. Some answers gave only a verbal description ("it's flat there") without the numeric value.

Solution

Solution. $\hat{S}(t)$ is a **right-continuous step function**: it only changes at observed event times and stays flat in between. To evaluate it at $t = 17$, find the **most recent event time at or before 17**, which is $t = 14$ (the subsequent $t = 15^+$ is a censoring, not an event, so the curve does not step there). So:

$$\hat{\mathrm{S}}_{\mathrm{KM}}(17) = \hat{\mathrm{S}}_{\mathrm{KM}}(14) = 0.714, \qquad \hat{\mathrm{S}}_{\mathrm{NA}}(17) = \hat{\mathrm{S}}_{\mathrm{NA}}(14) = 0.734$$

Give the actual numbers, not just "it's flat."

2.4 Median survival time is read off the curve, not modeled

Exam reference: ***Exercise 1.4*** *(1 point) — the KM and NA estimates of median survival time[5].*

⚠ Common mistake

Estimating the median by fitting an exponential[a] model ($\hat{\lambda}$ = events/total follow-up, then $\hat{\mathrm{E}}[T] = 1/\hat{\lambda}$). That computes an exponential-model **mean**, not the nonparametric **median**, and answers a different question. A second error was reading the median off the wrong curve or the wrong row.

[a] https://morrison-lab.github.io/pds/random-variables.html#def-exponential

Solution

Solution. The median survival time is the smallest t at which the estimated survival curve drops to (or below) 0.5:

$$\hat{t}_{\mathrm{median}} = \min\{t : \hat{S}(t) \le 0.5\}$$

Reading from Table 1:

- $\hat{\mathrm{S}}_{\mathrm{KM}}(t)$ first reaches ≤ 0.5 at $t = 18$ (where it drops to 0.476), so the KM median is **18**

[5] intro-to-survival-analysis.qmd#def-median-survival

months.
- $\hat{S}_{NA}(t)$ first reaches ≤ 0.5 at $t = 21$ (where it drops to 0.319), so the NA median is **21 months**.

We can confirm this in R:

```
quantile(KM_est, p = 0.5)$quantile
#> 50
#> 18
```

No distributional assumption (exponential or otherwise) is used: the Kaplan–Meier estimator[a] is a nonparametric read-off of the estimated curve.

[a] intro-to-survival-analysis.qmd#def-KM-estimator

2.5 How censored times are used: denominator yes, numerator no

Exam reference: ***Exercise 1.5*** *(2 points) — describe how the censored time(s) are utilized in the Kaplan-Meier estimate.*

⚠ Common mistake

Describing only the *symptom* ("the curve doesn't drop at a censoring time") without the *mechanism*, or implying that censored subjects are simply dropped from the analysis entirely.

Solution

Solution. A censored subject contributes to the **risk-set**[a] **denominators** (n_j) for every event time up to and including their censoring time, because we know they survived event-free until then. They **never** contribute to an event **numerator** (d_j), because no event was observed for them.
In this example, subjects #3 and #4 (censored at 14^+ and 15^+) are counted in the number at risk at $t = 10$ and $t = 14$, which is why those denominators are 7 and 6. After their censoring times they drop out of all later risk sets. This mechanism is exactly what lets Kaplan-Meier[b] use partial information from censored subjects instead of discarding them.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] intro-to-survival-analysis.qmd#def-KM-estimator

3 Part 2: Cox proportional hazards models

For reference, here is the fitted model from the exam. The primary exposure is baseline smoking category (nonsmoker = reference, light smoker $L = 1$, heavy smoker $H = 1$), adjusting for age in decades (A), behavior pattern (P), overweight (B), and high cholesterol (C).

Table 2: Cox PH regression for the WCGS CHD data

Characteristic	$\log(\widehat{HR})$	$\widehat{HR}$	95% CI for HR	p
nonsmoker (ref.)	–	–	–	–
light smoker (L)	0.36	1.43	(1.05, 1.95)	0.023
heavy smoker (H)	0.78	2.18	(1.63, 2.91)	$<$ 0.001
age per decade (A)	0.66	1.94	(1.56, 2.40)	$<$ 0.001
behavior type A (P)	0.73	2.06	(1.58, 2.70)	$<$ 0.001
overweight (B)	0.30	1.34	(1.05, 1.72)	0.019

Characteristic	$\log(\widehat{HR})$	$\widehat{HR}$	95% CI for HR	p
high cholesterol (C)	0.75	2.12	(1.65, 2.71)	$<$ 0.001

The exam also provided the estimated covariance matrix[6] of the coefficient estimates (on the $\log(HR)$ scale). The relevant entries are:

$$\mathrm{Var}(\hat{\beta}_L) = 0.024840, \quad \mathrm{Var}(\hat{\beta}_H) = 0.021905, \quad \mathrm{Cov}(\hat{\beta}_L, \hat{\beta}_H) = 0.010504$$

3.1 Writing the model: name every assumption and show where it is used

Exam reference: ***Exercise 2.1*** *(10 points) — write the mathematical form of the proportional hazards model corresponding to Table 1.*

⚠ Common mistake

Several patterns lost the most points here:

- Treating the question as "interpret the coefficients" and writing only the linear predictor[a], omitting the **likelihood**[b], the **distribution functions**, and the **assumptions**.
- Listing assumption *names* without **showing where they are used**, or writing the math without naming the assumption.
- Adding a spurious **"baseline hazard**[c] **is exponential"** assumption. The Cox model leaves the baseline hazard $\lambda_0(t)$ **unspecified**; assuming a parametric form is wrong.
- Omitting one or more of the standalone distribution functions (survival[d], density[e], hazard[f], log-hazard[g], cumulative hazard[h]).

[a] Intro-to-GLMs.qmd#def-linear-predictor
[b] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[c] proportional-hazards-models.qmd#def-base-hazard
[d] intro-to-survival-analysis.qmd#def-surv-fn
[e] https://morrison-lab.github.io/pds/random-variables.html#def-pdf
[f] intro-to-survival-analysis.qmd#def-hazard
[g] proportional-hazards-models.qmd#def-loghaz
[h] intro-to-survival-analysis.qmd#def-cuhaz

Solution

Solution. A full-credit answer connects the likelihood to the linear predictor through a chain of named components and assumptions. Using $\tilde{X} = (L, H, A, P, B, C)$:

Joint likelihood of the data set:

$$\mathcal{L} \overset{\text{def}}{=} \mathrm{p}(\tilde{Y} = \tilde{y}, \tilde{D} = \tilde{d} \mid \mathbf{X} = \mathbf{x})$$

Marginal likelihood contribution of observation i:

$$\mathcal{L}_i \overset{\text{def}}{=} \mathrm{p}(Y_i = y_i, D_i = d_i \mid \tilde{X}_i = \tilde{x}_i)$$

Independent*[a]*-observations assumption (used to factor the likelihood):

$$\mathcal{L} = \prod_{i=1}^{n} \mathcal{L}_i$$

Non-informative censoring*[b] *assumption (used so each contribution reduces to survival and hazard terms): $T_i \perp\!\!\!\perp C_i \mid \tilde{X}_i$, giving

$$\mathcal{L}_i \propto [\mathrm{f}(y_i \mid \tilde{x}_i)]^{d_i} [\mathrm{S}(y_i \mid \tilde{x}_i)]^{1-d_i} = \mathrm{S}(y_i \mid \tilde{x}_i) \cdot [\lambda(y_i \mid \tilde{x}_i)]^{d_i}$$

[6] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov-vec-x

Distribution functions (define each one):

$$\mathrm{S}(t \mid \tilde{x}) \stackrel{\text{def}}{=} \mathbb{P}(T > t \mid \tilde{X} = \tilde{x}) = \exp\{-\Lambda(t \mid \tilde{x})\}$$

$$\mathrm{f}(t \mid \tilde{x}) \stackrel{\text{def}}{=} \lambda(t \mid \tilde{x})\ \mathrm{S}(t \mid \tilde{x})$$

$$\lambda(t \mid \tilde{x}) \stackrel{\text{def}}{=} \mathrm{p}(T = t \mid T \geq t, \tilde{X} = \tilde{x}) = \frac{\mathrm{f}(t \mid \tilde{x})}{\mathrm{S}(t \mid \tilde{x})}$$

$$\Lambda(t \mid \tilde{x}) \stackrel{\text{def}}{=} \int_0^t \lambda(u \mid \tilde{x})\, du = -\log \mathrm{S}(t \mid \tilde{x})$$

$$\eta(t \mid \tilde{x}) \stackrel{\text{def}}{=} \log \lambda(t \mid \tilde{x})$$

Proportional-hazards[c] ***assumption*** (used to split the hazard into a baseline that depends only on time and a factor that depends only on covariates):

$$\lambda(t \mid \tilde{x}) = \lambda_0(t) \cdot \theta(\tilde{x})$$

where $\lambda_0(t)$ is the **unspecified** baseline hazard.
Logarithmic-link[d] ***assumption*** (used to make the covariate factor a function of a linear predictor):

$$\eta(t \mid \tilde{x}) = \eta_0(t) + \Delta\eta(\tilde{x}), \qquad \theta(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$$

Linear functional-form assumption (used to write the covariate term as a linear combination[e]):

$$\Delta\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} = \beta_L\, l + \beta_H\, h + \beta_A\, a + \beta_P\, p + \beta_B\, b + \beta_C\, c$$

Notice that the baseline hazard $\lambda_0(t)$ is carried along symbolically the whole time — we never assume a shape for it.

[a] https://morrison-lab.github.io/pds/independence.html#def-indpt
[b] intro-to-survival-analysis.qmd#def-noninformative-censoring
[c] proportional-hazards-models.qmd#def-pha
[d] Intro-to-GLMs.qmd#sec-why-link
[e] https://morrison-lab.github.io/mds/linear-algebra-vectors.html#def-linear-combination

3.2 Interpreting a hazard ratio: magnitude, reference, adjustment, *and* significance

Exam reference: ***Exercise 2.2*** *(4 points) — summarize how baseline smoking category is associated with hazard of incident CHD.*

⚠ Common mistake

- Reporting the direction of the effect but omitting the **magnitude** ("X% higher/lower hazard") or the **reference group**.
- Forgetting to say the estimate is **adjusted for / holding constant** the other covariates.
- Omitting **statistical significance**[a] (whether the 95% CI[b] excludes 1, or $p < 0.05$).
- Calling the hazard ratio[c] a **risk** or an **odds ratio**[d]. It is a ratio of *hazards*, not of risks or odds[e].

[a] https://morrison-lab.github.io/sds/inference.html#def-statistically-significant
[b] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[c] intro-to-survival-analysis.qmd#def-hazard-ratio
[d] binary-outcome-associations.qmd#def-OR
[e] binary-outcome-associations.qmd#def-odds

Solution

Solution. A complete interpretation has four parts: magnitude, reference group[a], adjustment, and significance. For the smoking effect in Table 2:

> Adjusting for age, behavior pattern, overweight, and cholesterol, **light smokers** had an estimated hazard of incident CHD about **43% higher** than **nonsmokers** ($\widehat{HR} = 1.43$); this hazard ratio is statistically significant at the 0.05 level, since the 95% CI $(1.05, 1.95)$ excludes 1 ($p = 0.023$).
>
> **Heavy smokers** had an estimated hazard about **118% higher** (roughly double; $\widehat{HR} = 2.18$) than nonsmokers, also adjusting for the other covariates; this hazard ratio is highly significant ($p < 0.001$, CI $(1.63, 2.91)$).

Note "43% higher" comes from $1.43 - 1 = 0.43$, and "118% higher" from $2.18 - 1 = 1.18$. Interpret each smoking level **separately** against the reference, rather than lumping them together.

[a] https://morrison-lab.github.io/sds/data.html#def-reference-level

3.3 A hazard ratio for a non-unit change: exponentiate

Exam reference: ***Exercise 2.3*** *(2 points) — the hazard ratio associated with a 7.5-year increase in age.*

⚠ Common mistake

For the hazard ratio[a] associated with a **7.5-year** increase in age (with the table reporting the HR per **decade**), the errors were:

- Multiplying: $1.94 \times 0.75 = 1.46$. **Wrong** — the HR for a multi-unit change is *not* linear in the HR.
- Arithmetic slips that produced answers like 2.08.

[a] intro-to-survival-analysis.qmd#def-hazard-ratio

Solution

Solution. On the **log** scale, the effect is linear: a change of c units multiplies the log-hazard[a] by $c \cdot \beta$. So the hazard ratio for a c-unit change is

$$HR(c) = \exp\{c\,\beta\} = (\exp\{\beta\})^c = HR^c$$

A 7.5-year increase is $c = 0.75$ decades, and the per-decade $\widehat{HR} = 1.94$ (i.e. $\hat{\beta}_A = \log 1.94 = 0.66$):

$$\widehat{HR}(0.75) = 1.94^{0.75} = \exp\{0.75 \times 0.66\} = \exp\{0.495\} = 1.64$$

So a 7.5-year-older man has about **64% higher** estimated hazard of CHD, all else equal. The operation is **exponentiation** ($HR^{0.75}$), not multiplication.

[a] proportional-hazards-models.qmd#def-loghaz

3.4 Comparing two coefficients: variance of a *difference*

Exam reference: ***Exercise 2.4*** *(2 points) — test whether the hazard of incident CHD differs between heavy and light smokers; compute the z-statistic*[7] *and two-sided p-value*[8]*.*

[7] https://morrison-lab.github.io/sds/inference.html#def-test-statistic
[8] https://morrison-lab.github.io/sds/inference.html#def-p-value

⚠ Common mistake

This mistake was the **single most consequential Part 2 error**. To test $H_0 : \beta_H = \beta_L$, the standard error[a] of $\hat{\beta}_H - \hat{\beta}_L$ was computed incorrectly in several ways:

- Using a **single covariance**[b] **entry** $\text{Cov}(\hat{\beta}_L, \hat{\beta}_H)$ as the variance[c].
- **Adding** 2 Cov instead of **subtracting** it.
- Plugging in the **hazard ratios** (2.18, 1.43) instead of the **coefficients** (0.78, 0.36) in the numerator.
- Guessing a value for z because the variance-of-a-difference formula[d] was not on the formula sheet.

[a] https://morrison-lab.github.io/sds/estimation.html#def-SE
[b] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov
[c] https://morrison-lab.github.io/pds/variance-covariance.html#def-variance
[d] https://morrison-lab.github.io/pds/variance-covariance.html#thm-var-lincom

Solution

Solution. Work on the $\log(HR)$ (coefficient) scale, where the estimates are approximately normal[a]. The point estimate of the difference is

$$\widehat{\Delta} = \hat{\beta}_H - \hat{\beta}_L = 0.78 - 0.36 = 0.42$$

The variance of a **difference** of two estimates uses **all three** relevant entries of the covariance matrix[b]:

$$\text{Var}(\hat{\beta}_H - \hat{\beta}_L) = \text{Var}(\hat{\beta}_H) + \text{Var}(\hat{\beta}_L) - 2\ \text{Cov}(\hat{\beta}_H, \hat{\beta}_L)$$

The -2 Cov term is essential — its sign is **minus** for a difference. Plugging in:

```
var_H <- 0.021905
var_L <- 0.024840
cov_HL <- 0.010504

diff <- 0.78 - 0.36
var_diff <- var_H + var_L - 2 * cov_HL
se_diff <- sqrt(var_diff)

z <- diff / se_diff
p_value <- 2 * pnorm(-abs(z))

c(diff = diff, se = se_diff, z = z, p_value = p_value) |> round(4)
#>    diff      se       z p_value
#>  0.4200  0.1604  2.6180  0.0088
```

So $z = 0.42/0.160 = 2.62$, two-sided $p = 0.0088$. We **reject** H_0: the hazard of CHD differs significantly[c] between heavy and light smokers, holding the other covariates constant.

Note

Compare with the **naive** standard error that ignores the covariance, $\sqrt{\text{Var}(\hat{\beta}_H) + \text{Var}(\hat{\beta}_L)} = 0.216$. Because $\hat{\beta}_H$ and $\hat{\beta}_L$ are *positively* correlated[d], subtracting 2 Cov makes the correct SE (0.160) **smaller**, which sharpens the test. Using a single covariance cell as the variance gave a wildly wrong SE and an absurd z.

[a] https://morrison-lab.github.io/pds/random-variables.html#def-normal
[b] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov-vec-x
[c] https://morrison-lab.github.io/sds/inference.html#def-statistically-significant
[d] https://morrison-lab.github.io/pds/variance-covariance.html#def-correlation

3.5 A confidence interval for a hazard ratio: build it on the log scale, then exponentiate

Exam reference: ***Exercise 2.5*** *(2 points) — a 95% confidence interval*[9] *for the hazard ratio*[10] *comparing heavy to light smokers.*

⚠ Common mistake

- Forming the interval directly on the HR scale, $\widehat{HR} \pm 1.96 \cdot \text{SE}$ (e.g. 1.52 ± 0.31), instead of on the log scale.
- Building the log-scale interval correctly but **never exponentiating**, then reporting a log-scale interval as if it were the HR (sometimes even declaring "significant" while the reported interval contained 1).
- Reusing the wrong SE from the Wald test[a] in Section 3.4.

[a] https://morrison-lab.github.io/sds/intro-MLEs.html#def-wald-test

Solution

Solution. A hazard ratio is positive and its sampling distribution is skewed, so the symmetric normal approximation is applied to $\log(HR)$, and the endpoints are **then** exponentiated. For the heavy-vs-light comparison ($\widehat{\Delta} = 0.42$, SE $= 0.160$):

$$\text{log-scale CI:} \quad 0.42 \pm 1.96 \times 0.160 = (0.106,\ 0.734)$$

$$\text{HR CI:} \quad (e^{0.106},\ e^{0.734}) = (1.11,\ 2.08)$$

```
est <- 0.42
se  <- sqrt(0.021905 + 0.024840 - 2 * 0.010504)
exp(est + c(-1, 1) * 1.96 * se) |> round(2)
#> [1] 1.11 2.08
```

The point estimate is $\widehat{HR} = e^{0.42} = 1.52$, and the 95% CI $(1.11, 2.08)$ **excludes 1**, consistent with the significant Wald test in Section 3.4. Always exponentiate the endpoints, and check that your CI and your hypothesis test agree.

4 Summary of the most common errors

#	Exam Q	Topic	The fix
1	1.1	Risk set after censoring (Section 2.1)	Remove **everyone** who has an event *or* is censored from later risk sets.
2	1.1	Nelson-Aalen (Section 2.2)	Sum hazards d_j/n_j, then $\hat{S}_{\text{NA}} = \exp\{-\hat{\Lambda}\}$.
3	1.3	Reading $\hat{S}(t)$ (Section 2.3)	Step function: use the most recent event time $\le t$; give the number.
4	1.4	Median survival (Section 2.4)	First t with $\hat{S}(t) \le 0.5$ — not a mean, not an exponential fit.
5	1.5	Role of censoring (Section 2.5)	Denominator until censoring; never the event numerator.

[9] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[10] intro-to-survival-analysis.qmd#def-hazard-ratio

#	Exam Q	Topic	The fix
6	2.1	Writing the model (Section 3.1)	Name **and** apply every assumption; baseline hazard stays unspecified.
7	2.2	Interpreting an HR (Section 3.2)	Magnitude + reference + adjustment + significance; HR $\neq$ risk/odds.
8	2.3	HR for a non-unit change (Section 3.3)	$HR^c = \exp\{c\beta\}$ — exponentiate, don't multiply.
9	2.4	Comparing two coefficients (Section 3.4)	$\text{Var}(\hat{\beta}_H - \hat{\beta}_L) = \text{Var}_H + \text{Var}_L - 2\ \text{Cov}$.
10	2.5	CI for an HR (Section 3.5)	Build on the log scale, then exponentiate the endpoints.